L4 Safety Island for Autoware — Implementation and Demo Plan

NXP S32Z2, Least-Cost branch • phases, work packages, partitions and issues

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National Taiwan University • Reference Design WG • Design Release V1.0 (20261023) • plan of 2026-11-13

Where everything is

Every slide in this deck points back to one of these

Document	What it holds	Link
Implementation repository	Plans, issues, progress counts	L4SafetyIsland_Implementation
Implementation plan, S32Z2	Nodes, dependency order, work packages, tests, demos, decisions	docs/PLAN-S32Z2.md
Milestones	M0–M7: phase reached and demo shown, in build order	docs/MILESTONES.md
WP0–WP3 node by node	Messages, WP1 partition map, the safety-floor nodes, demo D1	docs/WP0-WP3-SafetyFloor.md
Architecture	One code base for S32Z2 and R-Car X5H	docs/ARCHITECTURE.md
Issues	Tracking issue: pickup order, open decisions; all issues by WP and partition	#1 • all issues
Design, Release V1.0	The design this plan implements (version 20261023)	release v1.0
Design document	On GitHub; the §-references in this deck point here	design document

Takeaway. The plan documents are the source of truth. This deck summarises them; where they differ, the documents win.

What is being built

The Least-Cost branch of design D10 / §9.5, on the board in the lab

The S32Z2 island

Board	X-S32Z27X-DC, **FreeRTOS + nano-ros** (already evaluated)
Processor	8 × Cortex-R52 (lockstep / split), DSP, 19 MB SRAM
Camera (rank 1)	1 mono, 100BASE-T1 — **mandatory**
Radar (rank 2)	none → N6 not built
Ultrasonic (rank 3)	5
IMU / GNSS (rank 4)	1
Lidar (rank 5)	solid-state **3D LiDAR** in place of the 2D lidar; confirming only
Island ODD	city road, ~30 km/h, fair weather
Vehicle	NTU golf cart

Node count on this platform

Migrated Autoware nodes (`I*`) `I15` not needed on the golf cart	**17**
New island nodes (`N*`) `N6` absent, `N20` optional	**21**
New HPC nodes (`H*`), on Autoware	**8** + 1 optional

Code

New `N*` nodes in `no_std` Rust (Q2); migrated `I*` nodes stay C++ ports (Q8, proposed). One code base with the R-Car X5H, $\geq$ 85 % shared.

PLAN-S32Z2.md §1 • ARCHITECTURE.md

Phases

A phase is a capability the island reaches. A work package is the unit of work that gets it there.



The phases, defined

Design §12: each phase adds a set of nodes and reaches one capability

Phase	Partition	Adds	Capability reached	Work packages
0 — done	P3	I9–I12	Trajectory following on the island, closed loop against unmodified Autoware	WP1 (onto the board)
1	P0	I1–I3, I5–I7, I16, N1–N4, N8	**Autonomous fallback**: authority transfer on HPC loss; blind but bounded MRM	WP2, WP3
2	P1	I8, I13, N5, N7, N9, N10, N12	**Guardian role**: command validation, safety envelope, veto, two-tier control	WP4, WP7 (N9, N12 in WP2)
3	P2	I19, I20, N13–N17 (no N6)	**Obstacle-aware MRM**: the island stops for what it can see	WP5
4	P1	N18 + H8	**Pull-over MRM**: the island moves the vehicle out of the running lane	WP6
5	P2	N19	**ODD degradation**: independent visibility and friction verdict	WP7
added by V1.0		B2 hold (N5, v_hold) • B6 / B5 (N21–N23) • audit (N11, N20)	Lane hold after HPC loss; joystick and tele-operation; event recording	WP9, WP8, WP10

Takeaway. Phases 1 and 2 are the critical path. The design's shortest credible path to an L4 fallback is phases 1, 3 and 4 reduced to their mandatory nodes: a pull-over island in 21 nodes.

How the order was derived

From the edge catalogue of the message-flow diagram, restricted to this platform

Five rules

1. **Messages first.** Every edge is a typed topic, so `autoware_safety_island_msgs` and the E2E profiles come before any node (WP0).
2. **Hubs interface-first.** `I1` has 12 command sources and N8 11 fault sources; both are built early with every port declared.
3. **The B4 floor first.** Nine nodes — `N4 N3 N8 N17 I7 I6 I1 I16 I5` — read no exteroceptive topic; they are built and measured before anything above them (WP2–WP3).
4. **Sensing in parallel.** `N13`–`N17` have no dependency on P0 (WP5).
5. **Behaviours in dependency order**, not number order: **B4** → **B3** → **B1** (with the guardian) → B6/B5 → B2. B3 needs no N5 (§9.4); B2 needs it.

Dependency levels (forward edges only)

L0	I16 N12 N13 N14 N15 N17 N23
L1	I5 N1 N9 N16
L2	N2 N4
L3	I9(–I12) I19 I20 N5 N19 N21 N22
L4	I8 I14 N10
L5	N8 (11 fault producers)
L6	N3
L7	I2 I7 N11
L8	I3 I6 N18
L9	N7
L10	I13
L11	I1 (12 command sources)

PLAN-S32Z2.md §2

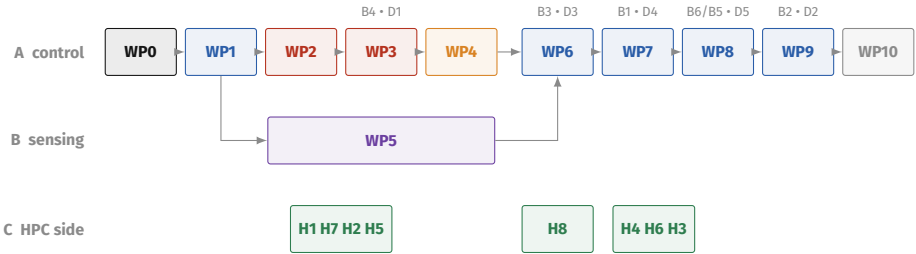
Work packages

Eleven work packages on three tracks. Every one ends in a test, a demo or both.



Sequence and tracks

Authority and control on the critical path; sensing and the HPC side in parallel



- The HPC-side nodes are **dependencies** of the island WPs: H1 H7 H2 H5 before WP2, H8 before WP6, H4 H6 H3 before WP7.
- **Minimal pull-over island** (design §9.4): WP2 + WP3 + WP4 + WP5 + WP6 deliver D1 and D3; the guardian role and B1 (WP7) follow, then B6/B5 (WP8); the B2 lane hold (WP9) comes last.

PLAN-S32Z2.md §3.1, §3.3, §3.4

The work packages at a glance

WP	Name	Builds	Exit / demo
WP0	Foundations	Messages with fixed capacity; SI-E2E/A and /B as nano-ros E2E configuration; contract files; test harness	Contracts clean in CI; E2E passes the §8.4 threat table
WP1	Phase 0 on the board	Partition map (synthetic load where no node yet); Cyclone DDS to the HPC; tracing; HIL rig; I9–I12 retargeted	Phase 0 reproduced on the board
WP2	Base layer L0–L2	I16 N12 N1 N4 N2 N9 I5	HPC liveness and ego state seen; faults reach N8
WP3	Safety floor and B4	N8 N3 I2 I3 I7 I6 I1, reduced N17	**D1**: HPC loss and blindness on the bench, ≤ 70 ms
WP4	Fallback control tier	N7 ramp, I13, I11 as MRM law, (I14)	In-lane bounded stop after HPC loss
WP5	Island sensing	N14 N13 N15 N17 N16 I19 I20	Obstacle-aware stop
WP6	B3 pull over	N18, island-detected refuge, lateral path; HPC loss → B3 directly	**D3**: the golf cart pulls over
WP7	Guardian and B1	I8 N5 N10 N19, compliance supervision in N2	**D4**: degraded visibility → ODD switch
WP8	B6 and B5	N23 wired joystick; N21 N22 as stubs (RCC deferred in V1.0)	**D5**: joystick drives the cart off the road
WP9	B2 lane hold	LANE_KEEP_HOLD, T_hold fork, v_hold, recovery verdict	**D2**: HPC fails, island holds the lane, HPC returns
WP10	Audit and options	N11 event recorder, N20 (optional)	Required before any pilot

PLAN-S32Z2.md §3.2

Milestones: a phase reached, and its demo shown

Pull-over island first, then the guardian and ODD degradation; lane hold last

Milestone	Capability (phase)	WPs	Demo / exit	Issues
M0	Foundations; phase 0 on the board	WP0, WP1	Phase 0 reproduced; contracts clean in CI	#2
M1	Safety floor (1)	WP2, WP3	**D1**: B4 on the bench, ≤ 70 ms	#3–#9
M2	Obstacle-aware bounded stop (3 + fallback tier)	WP4, WP5	In-lane stop, shortened for obstacles	#10 #12–#14
M3	Minimal pull-over island (4)	WP6	**D3**: the golf cart pulls over	—
M4	Guardian role and ODD degradation (2, 5)	WP7	**D4**: fogged lens $\rightarrow$ ODD switch verified	#11
M5	Local control (B6; B5 stubs)	WP8	**D5**: joystick drives the cart off the road	—
M6	Lane hold after HPC loss (B2)	WP9	**D2**: island holds the lane, HPC returns	—
M7	Pilot readiness (audit)	WP10	Event recorder; required before any pilot	—

- Until WP9, HPC loss forks straight to B3, and `N19`'s `T_odd` successor for HPC lost is B3: a staging of the design, not a change. WP9 restores B2 and re-runs D3 and D4.
- No due dates yet; they are set once WP0 has a measured pace.

MILESTONES.md • GitHub milestones

WP0–WP3: the safety floor, node by node

Detailed in WP0-WP3-SafetyFloor.md

WP0 — foundations

- 11 new message types, fixed capacity; internal types `FaultEvent`, `FaultVerdict`, `IslandMode`
- E2E: one Data ID per interface; profile A or B; §8.1 age limits
- Contracts: 70 ms = 30 detect + 20 decide + 20 command

WP1 — phase 0 on the board

- Six partitions at their §9.1 periods; synthetic load until nodes arrive
- Scheduling-matrix check; provisional intra-island transport (Q3)
- Tracing, HIL rig, host build from the same configuration
- `I9`–`I12` read Autoware directly until `N1`/`N4` exist

WP2 — base layer

- `N1`: every E2E failure is a fault event, never a silent drop
- `N2`: three missed heartbeats detected in ≤ 30 ms
- `N4`: island odometry, the odometry arbiter (Q6)

WP3 — authority core and B4

- `N8` and `I1` with **all** ports declared
- `N3`: `EMERGENCY_STOP` reachable from every state, no guard
- `I7`: availability from `N2`; 500 ms timeout re-parameterised
- `I7` $\leftrightarrow$ `I6` `OperateMrm` — phase-1 exit criterion

Takeaway. Exit of WP3: the nine-node B4 path works on the board, reads no exteroceptive topic, and is measured inside the 70 ms budget.

WP0-WP3-SafetyFloor.md • §2 WP1

Testing and demos

Test levels

Level	Where and what	From
T1 Unit	Host; node logic; Autoware's own suites for `I*`	WP0
T2 Node-in-the-loop	Host build; golden traces; contract terms	WP2
T3 SIL	Planning simulator as the HPC + host island	WP3
T4 HIL	X-S32Z27X-DC + HPC PC + CAN FD simulator	WP1
T5 Vehicle	Golf cart, closed course, ≤ 30 km/h	WP4

One board for all WPs: T2/T3 carry daily regression, T4 runs on merge and for timing.

Demos

Demo	WP	Scenario
D1	WP3	HPC loss, then camera unplugged: B4 on the bench (HIL)
D3	WP6	HPC does not return: the golf cart pulls over
D4	WP7	Fogged lens: ODD-switch demand, answer verified
D5	WP8	Joystick drives the stopped cart off the road
D2	WP9	HPC fails on a city road; island holds the lane; HPC returns

Takeaway. 15 injected faults — HPC hang, link flood, stale trajectory, E2E corruption, actuator non-response, every sensor lost — each with an expected response and the WP that verifies it.

PLAN-S32Z2.md §4

Partitions

Five executors with fixed priority bands. Under overload the island sheds the MPC, never the sensing.



Execution partitions on the S32Z2

Design §9.1, with this platform's node set and core placement

Partition	Tier	Nodes	Period	Band	S32Z2 placement
P0 safety core	T0	I1 I2 I3 I5 I6 I7 I16 N1 N2 N3 N4 N8	10 ms	1	R52 lockstep
P1 supervision + fallback	T1	I8 I13 N5 N7 N9 N10 N12 N18, N21 command path, N23	20 ms	2	R52 lockstep
P2 safety sensing	T1s	I19 I20 N13 N14 N15 N16 N17 N19 (N20)	20–50 ms	3	own core; N14 pixel stages on DSP / Neon
P3 nominal control	T2	I9 (hosting I10 I11 I12), I14	20–30 ms	4	R52 split
P3s tele-op streaming	T3	N22	33–100 ms	4b	R52 split; idle outside B5
P4 non-real-time	T3	N11, telemetry, N21 status uplink	100–1000 ms	5	R52 split

Rules the partitions encode

- **P0 must complete without P2 or P3**; it reads P2 only through an age-checked snapshot
- The QM nominal controller (P3) sits **below** ASIL-B sensing (P2): overload sheds the MPC, and I13 in P1 takes over within one cycle
- P4 is shed before anything else

How the plan checks them

- WP1: synthetic P0–P2 loads keep their periods under a P3 hog, an MPC overrun and an HPC link flood
- WP3 / WP4: repeated with the real nodes
- Open: intra-island transport for P0 (Q3, design open decision 7)

WP0-WP3-SafetyFloor.md §2.1 • ARCHITECTURE.md §4

Issues

One issue per imported Autoware node. The new island and HPC nodes are next.



Issues: the imported Autoware nodes

In dependency order; tracking issue #1 holds the pickup order and open decisions

Issue	Node	Package	WP	Part.
#1	—	**Tracking**: import the `I*` nodes	—	—
#2	`I9–I12`	trajectory follower (MPC + PID)	WP1	P3
#3	`I16`	vehicle interface (golf cart, CAN FD)	WP2	P0
#4	`I5`	stop_mode_operator	WP2	P0
#5	`I2+I4`	command_mode_decider + types	WP3	P0
#6	`I7`	mrm_handler	WP3	P0
#7	`I3`	command_mode_switcher	WP3	P0

Issue	Node	Package	WP	Part.
#8	`I6`	mrm_emergency_stop_operator	WP3	P0
#9	`I1`	control_command_gate	WP3	P0
#10	`I13`	pure_pursuit	WP4	P1
#11	`I8`	control_validator	WP7	P1
#12	`I14`	shift_decider (optional)	WP4	P3
#13	`I19`	autonomous_emergency_braking	WP5	P2
#14	`I20`	collision_detector	WP5	P2

Open issues by work package WP1 (1) • WP2 (2) • WP3 (5) • WP4 (2) • WP5 (2) • WP7 (1) • all P0 (7)

Takeaway. 14 issues, all open as of 2026-11-13. The README's progress counts are kept current by a GitHub workflow; the linked lists always show the live state.

README: Progress

What has no issue yet

New island nodes (N*) — 21 on this platform

- WP2: N1 N2 N4 N9 N12
- WP3: N3 N8, reduced N17
- WP4: N7
- WP5: N13 N14 N15 N16 N17
- WP6: N18 WP7: N5 N10 N19
- WP8: N21 N22 N23 WP10: N11 (N20)

Work-package-level items

- WP0: message package, E2E configuration, contracts, harness
- WP1: partition map, tracing, HIL rig

HPC-side nodes (H*), built on Autoware

- Before WP2: H1 E2E egress H7 heartbeat H2 capability H5 trajectory conditioner
- Before WP6: H8 refuge
- Before WP7: H4 ODD reporter H6 island state client H3 corridor

> Proposal: open the N* issues for WP0–WP3 next, with the same template as #2–#14 (inputs, outputs, tests, labels wpN and P0–P4).

Open decisions

Each is needed by a named work package

#	Question	Needed by
Q1	*Taken:* nano-ros built-in E2E satisfies SI-E2E/A and /B	—
Q2	*Taken:* Rust on the safety island for the new nodes	—
Q3	Intra-island transport for P0 (open decision 7): provisional in WP1, final before WP3 timing	WP1, WP3
Q4	Camera: classical CV or certified smart camera (open decision 8)	WP5
Q5	Blackout owner (`N17` or `N16`) and debounce (open decision 15)	WP3
Q6	Hazard-lamp owner, MRM longitudinal owner, odometry arbiter	WP2–WP4
Q7	Sensor placement on the golf cart, including the 3D LiDAR (open decision 21(c))	WP5
Q8	Migrated `I*` nodes stay C++ ports (proposed: yes)	WP0
Q9	The lab 3D LiDAR: model, interface, point rate, field of view, time sync	WP5

Takeaway. Q8 and Q3 block the first two work packages; Q5 blocks the exit of WP3. Those three are the decisions to take first.

PLAN-S32Z2.md §5